

Impermanent loss. Applies to *automated market makers* (AMM), where a contract holds assets on both sides of a trading pair. Suppose the AMM imposes a fixed exchange ratio between the two assets, and both assets appreciate in market value. The first asset appreciates by more than the second asset. Users drain the first asset, and the contract is left holding only the second asset. The impermanent loss is the value of the contract if no exchange took place (value of both tokens) minus the value of the contract after it was drained (value of second token).

Incentive. A broad term used to reward productive behavior. Examples include *direct incentives* and *staked incentives*.

Initial DeFi offering (IDO). A method of setting an initial exchange rate for a new token. A user can be the first liquidity provider on a pair, such as the new token and a *stablecoin* such as USDC. Essentially, the user establishes an artificial floor for the price of the new token.

Invariant. The result of a constant product rule. For example, $\text{invariant} = S_A \times S_B$, where S_A is the supply of asset A , and S_B is the supply of asset B . Suppose the instantaneous exchange rate is 1A:1B. The supply of asset $A = 4$ and the supply of asset $B = 4$. The invariant = 16. Suppose the investor wants to exchange some A for some B . The investor deposits 4 of A so that the contract has 8 A ($S_A = 4 + 4 = 8$). The investor can withdraw only 2 of asset B as defined by the invariant. The new supply of B is therefore 2 ($S_B = 4 - 2 = 2$).